Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Value of $\lim_{x\to 0}\frac{x\tan 2x-2x\tan x}{(1-\cos 2x)^2}$ is

(A) 2

(B) -2

(C) $1/2$

(D) $-1/2$

Q.2

If $\lim_{x\to 0}\frac{(\sin nx)[(a-n)nx-\tan x]}{x^2}=0$, then the value of a

(A) $\frac{1}{n}$

(B) $n-\frac{1}{n}$

(C) $n+\frac{1}{n}$

(D) None of these

Q.3

$\lim_{n\to\infty}\frac{5^{n+1}+3^n-2^{2n}}{5^n+2^n+3^{2n+3}}$ is equal to

(A) 5

(B) 3

(C) 2

(D) 0

Q.4

If $f(x)=\frac{\sin\left(e^{x-2}-1\right)}{\ln(x-1)}$, then $\lim_{x\to 2} f(x)$ is equal to

(A) -2

(B) -1

(C) 0

(D) 1

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Q.5

The value of $\lim_{x\to 0}\frac{(4^x-1)^3}{\sin\frac{x^2}{4}\log(1+3x)}$, is

(A) $\frac{4}{3}(\ln 4)^2$

(B) $\frac{4}{3}(\ln 4)^3$

(C) $\frac{3}{2}(\ln 4)^2$

(D) $\frac{3}{2}(\ln 4)^3$

Q.6

If $m, n \in I_0$ and $\lim_{x\to 0}\frac{\tan 2x - n\sin x}{x^3} =$ some integer, then value of this limit is

(A) 3

(B) 2

(C) $\frac{16+n}{12}$

(D) None of these

Q.7

$\lim_{x\to\infty}\left(\frac{x^{100}}{e^x}+\left(\cos\frac{2}{x}\right)^{x^2}\right) =$

(A) e^{-1}

(B) e^{-4}

(C) $\left(1+e^{-2}\right)$

(D) e^{-2}

Q.8

If $a = \min\left\{x^2+4x+5, x\in R\right\}$ and $b = \lim_{\theta\to 0}\frac{1-\cos 2\theta}{\theta^2}$, then the value of $\sum_{r=0}^{n} a^r\cdot b^{n-r}$ is

(A) $\frac{2^{n+1}-1}{4\cdot 2^n}$

(B) $2^{n+1}-1$

(C) $\frac{2^{n+1}-1}{3\cdot 2^n}$

(D) None of these

Q.9

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$\lim_{x\to\infty}\left(\frac{x^2}{3x-2}-\frac{x}{3}\right)=$

(A) $\frac{1}{3}$

(B) $\frac{2}{3}$

(C) $\frac{-2}{3}$

(D) $\frac{2}{9}$

Q.10

If $\lim_{x\to 0}\left(x^{-3}\sin 3x+ax^{-2}+b\right)$ exists and is equal to zero then:

(A) $\mathrm{a}=-3\&\ \mathrm{b}=9/2$

(B) $a=3\&b=9/2$

(C) $\mathbf{a}=-3\&\mathbf{b}=-9/2$

(D) $a=3\&b=-9/2$

Q.11

$\lim_{x\to 0}\frac{x}{\sqrt[3]{x^2+3x+8}-\sqrt[3]{x^2-5x+8}}$

(A) $\frac{2}{3}$

(B) $\frac{4}{3}$

(C) $\frac{3}{2}$

(D) $\frac{3}{4}$

Q.12

$\lim_{x\to 2}\frac{2^x+2^{3-x}-6}{\sqrt{2^{-x}}-2^{1-x}}$ is

(A) 2

(B) 4

(C) 8

(D) 16

Q.13

Let $f(x)=\lim_{n\to\infty}\frac{\log(2+x)-x^{2n}\sin x}{1+x^{2n}}$. Then

(A) $\lim_{x\to 1^+}f(x)\neq\lim_{x\to 1^-}f(x)$

(B) $\lim_{x\to 1^+}f(x)=\sin 1$

(C) $\lim_{x\to 1^-}f(x)$ doesn't exist

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(D) None of these

Q.14

If $f(x) = \begin{cases} \frac{[x]^2+\sin[x]}{[x]} & \text{for } [x] \neq 0, \text{ where } [x] \\ 0 & \text{for } [x] = 0 \end{cases}$

denotes the greatest integer less than or equal to x, then $\lim_{x\to 0} f(x)$ equals

(A) 1

(B) 0

(C) -1

(D) None of these

Q.15

Let $f(x) = x(-1)^{[1/x]}, x \neq 0$, where $[x]$ denotes
the greatest integer less than or equal to x then, $\lim_{x\to 0} f(x) =$

(A) does not exist

(B) 2

(C) 0

(D) -1

Q.16

If $A_i = \frac{x-a_i}{|x-a_i|}, i = 1, 2, 3, \ldots, n$ and
$a_1 < a_2 < a_3 \ldots . a_n$, then $\lim_{x\to a_m} (A_1 A_2 \ldots .. A_n), 1 \leq m \leq n$

(A) is equal to $(-1)^m$

(B) is equal to $(-1)^{m+1}$

(C) is equal to $(-1)^{m-1}$

(D) does not exist

Q.17

The value of $\lim_{x\to 1} \frac{\sqrt{1-\cos 2(x-1)}}{x-1}$ is

(A) exists and it equal $\sqrt{2}$

(B) exists and it equal $-\sqrt{2}$

(C) does not exist because $x - 1 \to 0$

(D) does not exist because the left hand limit is not equal to the right hand limit

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Q.18

If $\{x\}$ denotes the fractional part of x, then $\lim_{x \to [a]} \frac{e^{\{x\}} - \{x\} - 1}{\{x\}^2}$, where [a] denotes the integral part of a, is equal to

(A) 0

(B) $\frac{1}{2}$

(C) $e - 2$

(D) none of these

Q.19

$\lim_{x \to 0} \left[\min \left(y^2 - 4y + 11\right) \frac{\sin x}{x}\right]$ (where [ ] denotes the greatest integer function) is

(A) 5

(B) 6

(C) 7

(D) does not exist

Q.20

$\lim_{x \to 0} \left[\frac{\sin(\text{sgn}(x))}{(\text{sgn}(x))}\right]$, where [.] denotes the greatest integer function, is equal to

(A) 0

(B) 1

(C) -1

(D) does not exist

Answers & Solutions

Q.1 Answer:

$1/2$

Solution:

Consider $\lim_{x \to 0} \frac{x \tan 2x - 2x \tan x}{(1 - \cos 2x)^2}$

$= \lim_{x \to 0} \frac{x \tan 2x - 2x \tan x}{4 \sin^4 x} \quad \left(\because \cos 2x = 1 - 2 \sin^2 x\right)$

$= \lim_{x \to 0} \frac{x}{4 \sin^4 x} \left[\frac{2 \tan x}{1 - \tan^2 x} - 2 \tan x\right]$

$= \lim_{x \to 0} \frac{2x \tan x}{4 \sin^4 x} \left[\frac{1 - 1 + \tan^2 x}{1 - \tan^2 x}\right]$

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$= \lim_{x\to 0} \frac{2x\tan^3 x}{4\sin^4 x(1-\tan^2 x)}$

$= \frac{1}{2}\lim_{x\to 0} \frac{x}{\sin x}\cdot\frac{1}{\cos^3 x}\cdot\frac{1}{1-\tan^2 x} = \frac{1}{2}\cdot 1\cdot\frac{1}{1^3}\cdot\frac{1}{1-0} = \frac{1}{2}$

Q.2 Answer:

$n+\frac{1}{n}$

Solution:

Let $\lim_{x\to 0}\frac{(\sin nx)[(a-n)nx-\tan x]}{x^2} = 0$

$\Rightarrow \lim_{x\to 0}\frac{\left(nx-\frac{n^3x^3}{3!}\right)\left[n(a-n)x-\left\{x+\frac{x^3}{3}+\ldots\ldots\right\}\right]}{x^2} = 0$

(By using expansion of $\sin x$ **and** $\tan x$**)** $\Rightarrow n^2(a-n)-n=0 \Rightarrow an-n^2-1=0$

$\Rightarrow a = \frac{n^2+1}{n} = n+\frac{1}{n}$

Q.3 Answer:

0

Solution:

$\lim_{n\to\infty}\frac{5^{n+1}+3^n-2^{2n}}{5^n+2^n+3^{2n+3}} = \lim_{n\to\infty}\frac{5\cdot 5^n+3^n-4^n}{5^n+2^n+27\cdot 9^n}$

$= \lim_{n\to\infty}\frac{5\cdot\frac{5^n}{9^n}+\frac{3^n}{9^n}-\frac{4^n}{9^n}}{\frac{5^n}{9^n}+\frac{2^n}{9^n}+27} = \frac{0+0-0}{0+0+27} = 0$

Q.4 Answer:

1

Solution:

$f(x) = \frac{\sin\left(e^{x-2}-1\right)}{\ln(x-1)}$

$\lim_{x\to 2}\frac{\sin\left(e^{x-2}-1\right)}{\ln(x-1)} = L$

It is $\frac{0}{0}$ **(undefined) condition so using L'hospital's rule** $\Rightarrow \mathrm{L} = \lim_{\mathrm{x}\to 2}\left[\frac{\left\{\sin\left(\mathrm{e}^{\mathrm{x}-2}-1\right)\right\}}{\{\ln(\mathrm{x}-1)\}}\right]$

$\Rightarrow \mathrm{L} = \lim_{\mathrm{x}\to 2}\frac{\cos\left(\mathrm{e}^{\mathrm{x}-2}-1\right)\cdot\mathrm{e}^{(\mathrm{x}-2)}}{1/(\mathrm{x}-1)}$

$\Rightarrow \mathrm{L} = \lim_{\mathrm{x}\to 2}\cos\left(\mathrm{e}^{2-2}-1\right)\mathrm{e}^{2-2}\cdot(2-1)$

$\Rightarrow \mathrm{L} = \cos(0)\mathrm{e}^0.1 \Rightarrow \mathrm{L} = 1$

Q.5 Answer:

$\frac{4}{3}(\ln 4)^3$

Solution:

$\lim_{x\to 0}\frac{(4^x-1)^3}{\sin\frac{x^2}{4}\log(1+3x)}$

$= \lim_{x\to 0}\frac{(4^x-1)^3}{x^3}\cdot\frac{(x/2)^2}{\sin x^2/4}\cdot\frac{3x}{\log(1+3x)}\cdot\frac{4}{3}$

$= \frac{4}{3}\left(\log_e 4\right)^3.1\cdot\log_e(e) = \frac{4}{3}\left(\log_e 4\right)^3.$

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Q.6 Answer:

3

Solution:

$\lim_{x\to 0}\frac{\tan 2x-n\sin x}{x^3}=I$

$=\lim_{x\to 0}\frac{2x+\frac{8x^3}{3!}\ldots\ldots-nx+\frac{nx^3}{3!}}{x^3}=I$

$=\lim_{x\to 0}\frac{(2-n)x+\left(\frac{16+n}{6}\right)x^3+\ldots}{x^3}=I$

$\Rightarrow n=3$

Q.7 Answer:

e^{-2}

Solution:

Consider $\lim_{x\to\infty}\left[\frac{x^{100}}{e^x}+\left(\cos\frac{2}{x}\right)^{x^2}\right]$

$=\lim_{x\to\infty}\frac{x^{100}}{e^x}+\lim_{x\to\infty}\left[\cos\left(\frac{2}{x}\right)\right]^{x^2}$

$=\lim_{x\to\infty}\frac{x^{100}}{e^x}=0$ **(Using** L' **Hospital's rule** $)$

and $\lim_{x\to\infty}\left(\cos\frac{2}{x}\right)^{x^2}$ **is of** (1^∞) **form**

$=e^{\lim_{x\to\infty}x^2\left(\cos\frac{2}{x}-1\right)}$

$=e^{\lim_{t\to 0}\frac{4}{t^2}(\cos t-1)}\quad\left(\text{ Put }\frac{2}{x}=t\Rightarrow x=\frac{2}{t}\right)$

$=e^{-\lim_{t\to 0}\left(\frac{1-\cos t}{t^2}\right)\cdot 4}=e^{-\lim\left(\frac{\sin t}{2t}\right)4}=e^{-2}$

Q.8 Answer:

$2^{n+1}-1$

Solution:

$x^2+4x+5=(x+2)^2+1\geq 1$. **So,** $a=1$

$b=\lim_{\theta\to 0}\frac{2\sin^2\theta}{\theta^2}=2$

$\sum_{r=0}^{n}a^r\cdot b^{n-r}=b^n+ab^{n-1}+a^2b^{n-2}+\ldots..+a^n$

$=\frac{b^n\left[1-\left(\frac{a}{b}\right)^{n+1}\right]}{1-\frac{a}{b}}=\frac{2^n\left[1-\left(\frac{1}{2}\right)^{n+1}\right]}{1-\frac{1}{2}}$

$=\frac{2^{n+1}\left(2^{n+1}-1\right)}{2^{n+1}}=\left(2^{n+1}-1\right)$

Q.9 Answer:

$\frac{2}{9}$

Solution:

Consider $\lim_{x\to\infty}\left[\frac{x^2}{3x-2}-\frac{x}{3}\right]$

$=\lim_{x\to\infty}\left[\frac{3x^2-x(3x-2)}{3(3x-2)}\right]=\lim_{x\to\infty}\frac{2x}{3(3x-2)}$

$=\lim_{x\to\infty}\frac{2}{3}\frac{1}{\left(3-\frac{2}{x}\right)}=\frac{2}{9}$

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Q.10 Answer:

$\mathrm{a}=-3 \& \mathrm{b}=9/2$

Solution:

a)

$\lim_{x\to 0}\frac{\sin 3x}{x^3}+\frac{a}{x^2}+b=\lim_{x\to 0}\frac{\sin 3x+ax+bx^3}{x^3}=\lim_{x\to 0}\frac{3\frac{\sin 3x}{3x}+a+bx^2}{x^2}$

for existence of limit $3+a=0\Rightarrow a=-3$

$\therefore l=\lim_{x\to 0}\frac{\sin 3x-3x+bx^3}{x^3}$

$=27\cdot\frac{\sin t-t}{t^3}+b=0\quad(3x=t)$ **[OR use** L **' Hospital's rule]**

$=-\frac{27}{6}+b=0\Rightarrow b=\frac{9}{2}$

Q.11 Answer:

$\frac{3}{2}$

Solution:

$\lim_{x\to 0}\frac{x}{\sqrt[3]{x^2+3x+8}-\sqrt[3]{x^2-5x+8}}$

$=\lim_{x\to 0}\frac{x\left[\left(x^2+3x+8\right)^{\frac{2}{3}}+\left(x^2-5x+8\right)^{\frac{2}{3}}+\left(x^2+3x+8\right)^{\frac{1}{3}}\left(x^2-5x+8\right)^{\frac{1}{3}}\right]}{(x^2+3x+8)-(x^2-5x+8)}$

$=\lim_{x\to 0}\frac{\left[\left(x^2+3x+8\right)^{\frac{2}{3}}+\left(x^2-5x+8\right)^{\frac{2}{3}}+\left(x^2+3x+8\right)^{\frac{1}{3}}\left(x^2-5x+8\right)^{\frac{1}{3}}\right]}{8}$

$=\frac{\left[8^{\frac{2}{3}}+8^{\frac{2}{3}}+8^{\frac{1}{3}}8^{\frac{1}{3}}\right]}{8}=\frac{12}{8}=\frac{3}{2}$

Q.12 Answer:

8

Solution:

$\lim_{x\to 2}\frac{2^x+2^{3-x}-6}{\sqrt{2^{-x}}-2^{1-x}}=\lim_{x\to 2}\frac{(2^x)^2-6.2^x+2^3}{\sqrt{2^x}-2}$

(Multiplying Num, and Den. by 2^x **)**

$\lim_{x\to 2}\frac{(2^x-4)(2^x-2)\left(\sqrt{2^x}+2\right)}{\left(\sqrt{2^x}-2\right)\left(\sqrt{2^x}+2\right)}$

$\lim_{x\to 2}\frac{(2^x-4)(2^x-2)\left(\sqrt{2^x}+2\right)}{\left(\sqrt{2^x}-2\right)\left(\sqrt{2^x}+2\right)}$

$\lim_{x\to\infty}\left(2^x-2\right)\left(\sqrt{2^x}+2\right)=\left(2^2-2\right)\left(\sqrt{2^2}+2\right)=8$

Q.13 Answer:

$\lim_{x\to 1^+}f(x)\neq\lim_{x\to 1^-}f(x)$

Solution:

For $|x|<1, x^{2n}\to 0$ **as** $n\to\infty$ **and**

For $|x|>1, \frac{1}{x^{2n}}\to 0$ **as** $n\to\infty$

So,

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$$f(x)=\begin{cases}\log(2+x) & \text{if } |x|<1\\ \lim_{n\to\infty}\frac{x^{-2n}\log(2+x)-\sin x}{x^{-2n}+1}=-\sin x & \text{if } |x|>1\\ \frac{1}{2}[\log(2+x)-\sin x] & \text{if } |x|=1\end{cases}$$

Thus $\lim_{x\to1^+}f(x)=\lim_{x\to1+}(-\sin x)=-\sin 1$

and $\lim_{x\to1^-}f(x)=\lim_{x\to1^-}\log(2+x)=\log 3$

Hence, L.H.L $\neq$ **R.H.L**

Q.14 Answer:

None of these

Solution:

As $x\to0-$ **(i.e., approaches 0 from the left),** $[x]=-1.\lim_{x\to0^-}f(x)=\lim_{x\to0^-}\frac{1+\sin(-1)}{-1}=-1+\sin 1$

Whereas, if $x\to0^+$ **we get** $[x]=0.\therefore f(x)=0\quad\Rightarrow\lim_{x\to0^+}f(x)=0$

Thus, $\lim_{x\to0}f(x)$ **does not exist.**

Q.15 Answer:

0

Solution:

$\because[1/x]=$ Integer $\quad\therefore(-1)^{[1/x]}=1$ or -1

$\lim\limits_{x\to0}x(-1)^{[1/x]}=\lim\limits_{h\to0}(h)(1\text{ or }-1)=0$

$\lim\limits_{h\to0}(-h)(1\text{ or }-1)=0$

Q.16 Answer:

does not exist

Solution:

$A_i=\frac{x-a_i}{|x-a_i|},i=1,2,3,\ldots,n$

$a_1<a_2<a_3<\ldots..<a_n$

If x **is in the left neighbourhood of** $a_1<a_2<\ldots.<a_{m-1}<x<a_m<a_{m+1}<\ldots<a_n$

$A_i=\frac{x-a_i}{x-a_i}=1,i=1,2,\ldots,m-1$

$A_i=\frac{x-a_i}{(a_i-x)}=-1$

$i=m,m-1,\ldots.,n$

$\therefore A_1A_2\ldots...A_n=(-1)^{n-m+1}$**...(i)**

If x **is in the right neighbourhood of** $a_m,a_1<a_2<\ldots<a_{m-1}<a_m<x<a_{m+1}<\ldots<a_n,$

$A_i=\frac{x-a_i}{x-a_i}=1,i=1,2,\ldots..,n$

$\therefore A_1A_2\ldots..A_n=(-1)^{n-m}$**...(ii)**

$\therefore\lim_{x\to a_m^-}(A_1A_2\ldots..A_n)=(-1)^{n-m+1}$

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and $\lim_{x\to a_m^+}(A_1A_2.....A_n) = (-1)^{n-m} \quad \therefore LHL \neq RHL$

Hence, $\lim_{x\to a_m}(A_1A_2\ldots A_n)$ **does not exist.**

Q.17 Answer:

does not exist because the left hand limit is not equal to the right hand limit

Solution:

We have, $1-\cos 2(x-1) = 2\sin^2(x-1)$

$\Rightarrow \sqrt{1-\cos 2(x-1)} = \sqrt{2}|\sin(x-1)|$

Now, $\lim_{x\to 1^-}\frac{\sqrt{1-\cos 2(x-1)}}{x-1}$

$= \lim_{x\to 1^-}\frac{\sqrt{2}|\sin(x-1)|}{x-1} = \lim_{h\to 0}\frac{\sqrt{2}|\sin(1-h-1)|}{1-h-1} = \sqrt{2}\lim_{h\to 0}\frac{|\sin h|}{-h} = -\sqrt{2}\lim_{h\to 0}\frac{\sin h}{h}$

$= -\sqrt{2}(1) = -\sqrt{2}$

and $\lim_{x\to 1^-}\frac{\sqrt{1-\cos 2(x-1)}}{x-1}$

$= \lim_{x\to 1^-}\frac{\sqrt{2}|\sin(x-1)|}{x-1} = \lim_{h\to 0}\frac{\sqrt{2}|\sin(1+h-1)|}{1+h-1} = \sqrt{2}\lim_{b\to 0^+}\frac{|\sin h|}{h}$

$= \sqrt{2}\lim_{h\to 0}\frac{\sin h}{h} = \sqrt{2}(1) = \sqrt{2}$

$\therefore$ **left hand limit** $\neq$ **right hand limit.**

Q.18 Answer:

none of these

Solution:

Let [a] = n**, then** $\lim_{\mathrm{x}\to \mathrm{n}^-}\frac{\mathrm{e}^{\{\mathrm{x}\}}-\{\mathrm{x}\}-1}{\{\mathrm{x}\}^2}$

$= \lim_{h\to 0}\frac{e^{\{n-h\}}-\{n-h\}-1}{\{n-h\}^2} = \lim_{h\to 0}\frac{e^{1-h}-(1-h)-1}{(1-h)^2}$

$= \mathrm{e}-2$

and $\lim_{x\to n^+}\frac{e^{\{x\}}-\{x\}-1}{\{x\}^2}$

$= \lim_{\mathrm{h}\to 0}\frac{\mathrm{e}^{\{\mathrm{n+h}\}}-\{\mathrm{n+h}\}-1}{\{\mathrm{n+h}\}^2}$

$= \lim_{\mathrm{h}\to 0}\frac{\mathrm{e}^{\mathrm{h}}-\mathrm{h}-1}{\mathrm{h}^2}$

$= \lim_{\mathrm{h}\to 0}\frac{1+\mathrm{h}+\frac{\mathrm{h}^2}{2!}+\frac{\mathrm{h}^3}{3!}+\ldots.-\mathrm{h}-1}{\mathrm{h}^2} = \frac{1}{2}$

- Limit does not exist.

Q.19 Answer:

6

Solution:

$\min\left(y^2-4y+11\right) = \min\left[(y-2)^2+7\right] = 7$

or $\quad L = \lim_{x\to 0}\left[\min\left(y^2-4y+11\right)\frac{\sin x}{x}\right]$

$= \lim_{x\to 0}\left[\frac{7\sin x}{x}\right]$

$= [$ **a value slightly lesser than 7** $(|\sin x| < |x|$**, when** $x\to 0)$

$= \lim_{x\to 0}\left[7\frac{\sin x}{x}\right] = 6$

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Q.20 Answer:

0

Solution:

$\lim_{x \to 0+} \left[\frac{\sin(\operatorname{sgn} x)}{\operatorname{sgn}(x)}\right] = \lim_{x \to 0+} \left[\frac{\sin 1}{1}\right] = 0$

$= \lim_{x \to 0^-} \left[\frac{\sin(\operatorname{sgn} x)}{\operatorname{sgn}(x)}\right] = \lim_{x \to 0^-} \left[\frac{\sin(-1)}{-1}\right]$

$= \lim_{x \to 0^-} [\sin 1]$

Hence, the given limit is 0 .

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